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EXP 7 COMPUTERNETWORK

AIM: To implement a sliding window flow control protocol in C, where the sender transmits frames within a window through Sender_Buffer.txt and the receiver replies with ACK or NACK through Receiver_Buffer.txt.

Sender.c:

```
#include   <stdio.h>
#include  <stdlib.h>
#include <string.h>

#ifdef _WIN32
#include <windows.h>
void delay(int seconds)
{
   Sleep(seconds * 1000);
}
#else
#include <unistd.h>
void delay(int seconds)
{
   sleep(seconds);
}
#endif

#define SENDER_FILE "Sender_Buffer.txt"
#define RECEIVER_FILE "Receiver_Buffer.txt"

void clearFile(const char *filename)
{
   FILE *fp = fopen(filename, "w");

   if (fp != NULL)
      fclose(fp);
```

```
}

int readAcknowledgement(int *type, int *number)
{
   FILE *fp;
   char line[100];

   fp = fopen(RECEIVER_FILE, "r");

   if (fp == NULL)
      return 0;

   if (fgets(line, sizeof(line), fp) == NULL)
   {
      fclose(fp);
      return 0;
   }

   fclose(fp);

   if (sscanf(line, "ACK:%d", number) == 1)
   {
      *type = 1;
      return 1;
   }

   if (sscanf(line, "NACK:%d", number) == 1)
   {
      *type = 2;
      return 1;
   }
```

```
    return 0;
}

void writeFrames(char *message, int start, int end)
{
    FILE *fp;

    fp = fopen(SENDER_FILE, "w");

    if (fp == NULL)
    {
        printf("Error: Cannot open Sender_Buffer.txt\n");
        exit(1);
    }

    printf("\n--------------------------\n");
    printf("Sending frames %d to %d\n", start, end);
    printf("--------------------------\n");

    for (int i = start; i <= end; i++)
    {
        fprintf(fp, "Frame:%d DATA:%c\n", i, message[i]);

        printf("Frame No: %d   DATA: %c\n",
               i,
               message[i]);
    }

    fclose(fp);
}
```

```
int main()
{
    char message[500];
    int windowSize;

    int totalFrames;
    int base = 0;

    int ackType;
    int ackNumber;

    printf("========== SLIDING WINDOW SENDER ==========\n\n");

    printf("Enter window size: ");
    scanf("%d", &windowSize);

    if (windowSize <= 0)
    {
        printf("Invalid window size.\n");
        return 1;
    }

    getchar();

    printf("Enter text message: ");
    fgets(message, sizeof(message), stdin);

    message[strcspn(message, "\n")] = '\0';

    totalFrames = strlen(message);
```

```
if (totalFrames == 0)
{
   printf("Message cannot be empty.\n");
   return 1;
}

clearFile(SENDER_FILE);
clearFile(RECEIVER_FILE);

printf("\nMessage: %s\n", message);
printf("Total frames: %d\n", totalFrames);
printf("Window size: %d\n", windowSize);

while (base < totalFrames)
{
   int end = base + windowSize - 1;

   if (end >= totalFrames)
      end = totalFrames - 1;

   writeFrames(message, base, end);

   printf("\nSender_Buffer.txt updated.");
   printf("\nWaiting for acknowledgement");

   while (1)
   {
      delay(2);

      if (readAcknowledgement(&ackType, &ackNumber))
```

```
    {
        if (ackType == 1)
        {
            printf("\nReceived ACK:%d\n", ackNumber);

            if (ackNumber >= base)
            {
                base = ackNumber + 1;

                clearFile(RECEIVER_FILE);

                printf("Sliding window moved.\n");
                break;
            }
        }
        else if (ackType == 2)
        {
            printf("\nReceived NACK:%d\n", ackNumber);

            printf("Resending frames from frame %d...\n",
                  ackNumber);

            clearFile(RECEIVER_FILE);

            if (ackNumber >= base)
                base = ackNumber;

            break;
        }
    }
}
```

```
    }

    clearFile(SENDER_FILE);

    printf("\n=======================================\n");
    printf("All frames transmitted successfully!\n");
    printf("Message transmission completed.\n");
    printf("=======================================\n");

    return 0;
}
```

Receiver.c:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#ifdef _WIN32
#include <windows.h>
void delay(int seconds)
{
    Sleep(seconds * 1000);
}
#else
#include <unistd.h>
void delay(int seconds)
{
    sleep(seconds);
}
```

```
#endif

#define SENDER_FILE "Sender_Buffer.txt"
#define RECEIVER_FILE "Receiver_Buffer.txt"

int readFrames(int *frameNumbers, char *data, int maxFrames)
{
   FILE *fp;
   char line[200];

   int count = 0;

   fp = fopen(SENDER_FILE, "r");

   if (fp == NULL)
      return 0;

   while (fgets(line, sizeof(line), fp) != NULL &&
         count < maxFrames)
   {
      int frameNo;
      char ch;

      if (sscanf(line, "Frame:%d DATA:%c",
               &frameNo, &ch) == 2)
      {
         frameNumbers[count] = frameNo;
         data[count] = ch;

         count++;
      }
```

```
    }

    fclose(fp);

    return count;
}

void writeACK(int number)
{
    FILE *fp = fopen(RECEIVER_FILE, "w");

    if (fp == NULL)
        return;

    fprintf(fp, "ACK:%d\n", number);

    fclose(fp);
}

void writeNACK(int number)
{
    FILE *fp = fopen(RECEIVER_FILE, "w");

    if (fp == NULL)
        return;

    fprintf(fp, "NACK:%d\n", number);

    fclose(fp);
}
```

```
int main()
{
   int expectedFrame = 0;

   int frameNumbers[100];
   char data[100];

   int lastFirstFrame = -1;

   printf("========== SLIDING WINDOW RECEIVER ==========\n\n");

   printf("Receiver is waiting for frames...\n");

   while (1)
   {
      int count = readFrames(frameNumbers, data, 100);

      if (count > 0)
      {
         /*
          * Process only a new Sender_Buffer window.
          */
         if (frameNumbers[0] != lastFirstFrame)
         {
            int valid = 1;

            printf("\n-------------------------\n");
            printf("Frames received:\n");
            printf("-------------------------\n");

            for (int i = 0; i < count; i++)
```

```
{
   printf("Frame No: %d   DATA: %c\n",
         frameNumbers[i],
         data[i]);

   if (frameNumbers[i] != expectedFrame)
   {
      printf("\nExpected Frame: %d\n",
            expectedFrame);

      printf("Received Frame: %d\n",
            frameNumbers[i]);

      printf("ERROR DETECTED!\n");

      printf("Sending NACK:%d\n",
            expectedFrame);

      writeNACK(expectedFrame);

      valid = 0;
      break;
   }

   expectedFrame++;
}

if (valid)
{
   int ackNumber = expectedFrame - 1;
```

```
                printf("\nAll frames received correctly.\n");

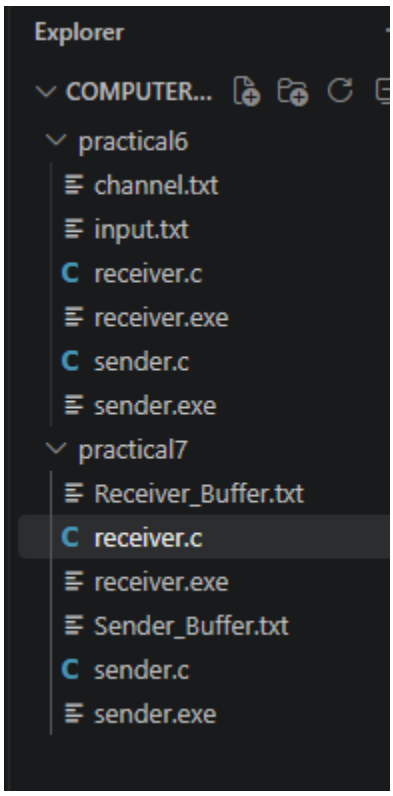
                printf("Sending ACK:%d\n",
                    ackNumber);

                writeACK(ackNumber);
            }

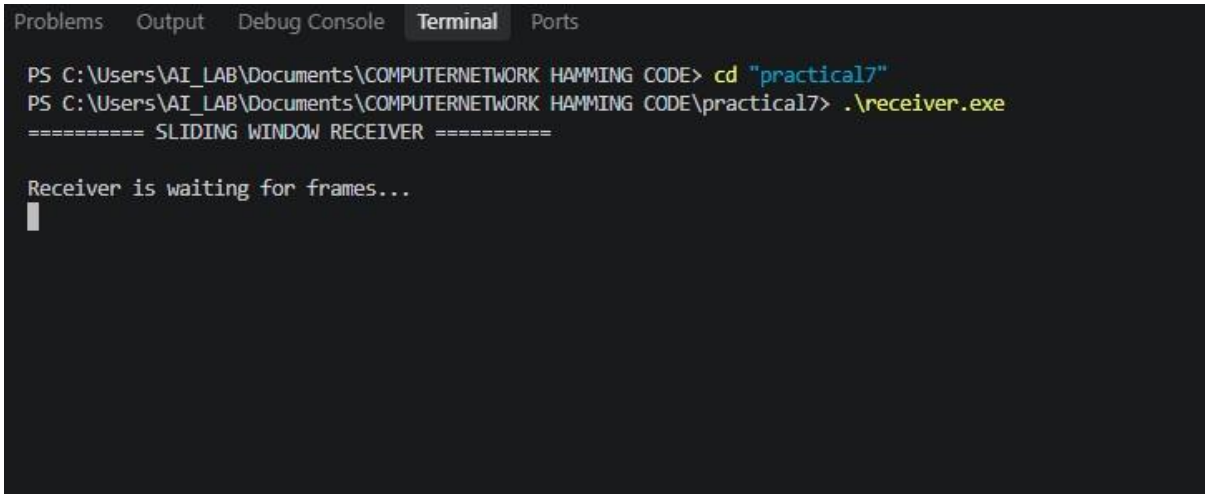
            lastFirstFrame = frameNumbers[0];
        }
    }

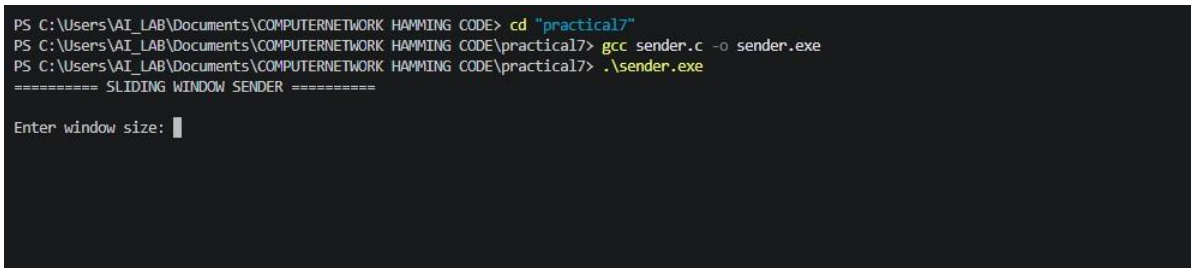
    delay(1);
  }

  return 0;
}
```



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Sender:

```
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE> cd "
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\pract
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\pract
========== SLIDING WINDOW SENDER ==========

Enter window size: 3
Enter text message: computer

Message: computer
Total frames: 8
Window size: 3

--------------------------------------
Sending frames 0 to 2
--------------------------------------
Frame No: 0   DATA: c
Frame No: 1   DATA: o
Frame No: 2   DATA: m

Sender_Buffer.txt updated.
Waiting for acknowledgement
Received ACK:2
Sliding window moved.
```

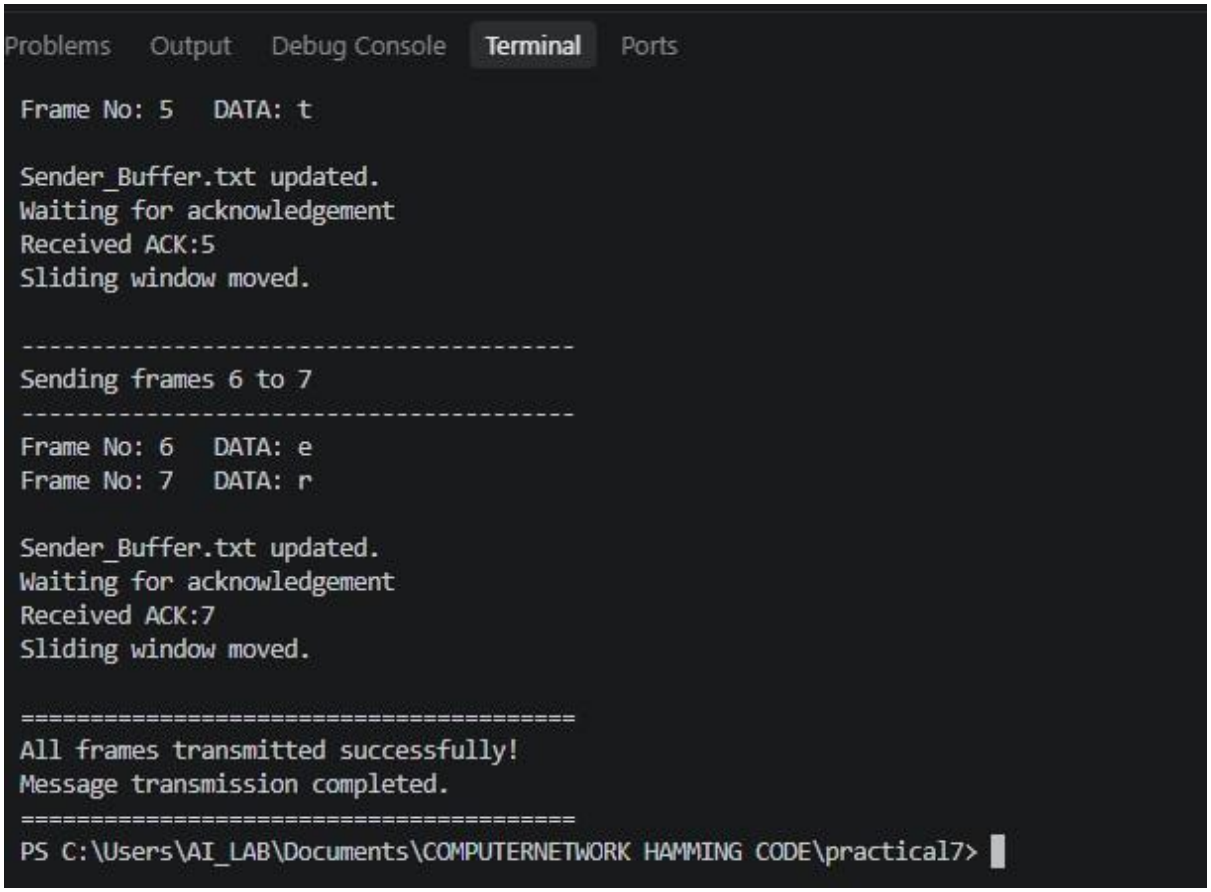
```
--------------------------------------
Sending frames 3 to 5
--------------------------------------
Frame No: 3   DATA: p
Frame No: 4   DATA: u
Frame No: 5   DATA: t

Sender_Buffer.txt updated.
Waiting for acknowledgement
Received ACK:5
Sliding window moved.

--------------------------------------
Sending frames 6 to 7
--------------------------------------
Frame No: 6   DATA: e
Frame No: 7   DATA: r

Sender_Buffer.txt updated.
Waiting for acknowledgement
Received ACK:7
Sliding window moved.
```

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Receiver:

```
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE> cd "practical7"
PS C:\Users\AI_LAB\Documents\COMPUTERNETWORK HAMMING CODE\practical7> .\receiver.exe
========== SLIDING WINDOW RECEIVER ==========

Receiver is waiting for frames...

---------------------------------------
Frames received:
---------------------------------------
Frame No: 0   DATA: c
Frame No: 1   DATA: o
Frame No: 2   DATA: m

All frames received correctly.
Sending ACK:2

---------------------------------------
Frames received:
---------------------------------------
Frame No: 3   DATA: p
Frame No: 4   DATA: u
Frame No: 5   DATA: t
```

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